

Week 04 Knowledge Check — TPM Academy

Week 4 Knowledge Check — Technical Architecture & Constraints

A short retention check covering the week's core ideas: monolith vs microservices and the three-question frame, STRIDE threat modeling and compliance, the C4 model, SLI/SLO/SLA and error budgets, and mature architectural trade-offs. Answer each question, then check yourself against the key at the end. Aim for concepts, not trivia — every answer maps back to something we used to build the TCD.

Format: 12 questions (multiple choice + true/false). ~15 minutes. Individual or triad.

Questions

1. (MC) A multi-deploy system whose services *must* all be released together to work is best described as a:

- A. Modular monolith
- B. Service-oriented architecture (SOA)
- C. Distributed monolith
- D. Event-driven architecture

2. (T/F) In the architecture stance template, writing “**Revisit if:** later / when needed” is an acceptable revisit trigger.

3. (MC) In a STRIDE pass, which letter is called out as the *most-missed* — the least intuitive one that teams building without audit trails routinely skip?

- A. Spoofing
- B. Repudiation
- C. Information disclosure
- D. Elevation of privilege

4. (T/F) A strong security brief opens by telling the security team “we don’t know anything yet, please tell us what to do.”

5. (MC) In the C4 model, which two levels are the TPM’s scope to draw?

- A. Context and Container
- B. Container and Component
- C. Component and Code
- D. Context and Component

6. (MC) When stress-testing a Container diagram with the three lenses (failure, trust boundary, evolvability), which annotation is called out as the *most-missed* — the place where compliance and security risk concentrates?

- A. The failure path
- B. The trust boundary
- C. The evolvability arrows
- D. The async/sync distinction

7. (T/F) Under a modular-monolith stance, the modules should be drawn inside one container as labeled sub-boxes, not promoted to separate containers.

8. (MC) Which of these is a *contractual commitment with consequences* — as opposed to an internal target?

- A. SLI
- B. SLO
- C. SLA
- D. Error budget

9. (MC) If a feature’s availability SLO is 99.9% over a 30-day window, its error budget is roughly:

- A. 4 minutes
- B. 43 minutes
- C. 7 hours
- D. 22 hours

10. (T/F) Stating a latency SLO as “avg < 500ms” is a solid target because the average captures typical user experience.

11. (MC) Which of these is a *mature* trade-off statement rather than a mere description?

- A. “We considered microservices but chose a monolith.”
- B. “We chose a monolith because monoliths are simpler.”
- C. “Option A: modular monolith. Option B: extract a service. We chose A. Accepted cost: future extraction needs a coordinated refactor. Revisit if traffic exceeds 5× rest-of-app.”
- D. “We’ll start with a monolith and figure out the rest later.”

12. (MC) When writing the top-5 architectural trade-offs, what is the recommended coverage check to confirm you surfaced real tensions?

- A. All 5 come from a single category
 - B. They span at least 3 of the five trade-off categories
 - C. Each names a specific competitor
 - D. Each has a dollar figure attached
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Answer Key

- 1. C** — Distributed monolith: the anti-pattern where you pay the ops cost of many deploys but still must release them together. “Microservices” in name only.
- 2. False** — “Later” or “when needed” is not a trigger. The senior-author signal is a concrete metric or org change (e.g., “if traffic grows 10x” or “if a second team contributes”).
- 3. B** — Repudiation. The least intuitive letter; teams that ship without audit trails miss it. A threat list with zero R’s should be pushed on.
- 4. False** — A passive “please tell us everything” brief signals an outsourced-thinking TPM. A strong brief is short, specific (concrete open questions), and confident (decisions already made, with justification).
- 5. A** — Context and Container. Stay above Component — the architect and engineers own the Component and Code layers.
- 6. B** — The trust boundary: where data crosses from our control to someone else’s. It’s the most-missed annotation and where security/compliance risk concentrates.
- 7. True** — A modular monolith is one container; draw the modules as sub-boxes inside it. Don’t promote them to separate containers — that would misrepresent the stance.
- 8. C** — SLA: a contractual commitment with consequences. SLI is the measurement, SLO is the internal target, and error budget is $(1 - \text{SLO})$ over the window. Calling an SLO an SLA casually is a common error.
- 9. B** — ~43 minutes. Error budget = $(1 - 0.999) = 0.1\%$ of 30 days ≈ 43 minutes of allowed downtime.
- 10. False** — Average hides bimodal distributions. Force percentiles (e.g., p95); every SLO needs a percentile, a window, and a defense.
- 11. C** — The mature pattern is Option A vs Option B $\rightarrow$ choice $\rightarrow$ accepted cost $\rightarrow$ revisit trigger. A, B, and D are descriptions, not trade-offs.
- 12. B** — Span at least 3 of the five categories (coupling vs independence, consistency vs availability, latency vs durability, speed vs robustness, generality vs simplicity). All in one category usually means real tensions were missed.